Journal of Synthetic Electrochemistry 12 (2026) 100042

Interphase repair, not particle fracture, limits silicon anode life

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Abstract Silicon-graphite composite anodes lose capacity faster than graphite alone, and the usual explanation blames fracture of the active particles. We show instead that the dominant loss channel is repeated repair of the solid electrolyte interphase after volume-driven cracking. Cells cycled 500 times at 0.5 C retained 82.1 +/- 1.4 % of their initial capacity, corresponding to a mean loss of 0.0837 % per cycle ($p < 0.001$).

Keywords: lithium-ion, silicon anode, interphase, capacity fade

1. Introduction

Lithium-ion cells with silicon-bearing anodes promise higher energy density, but they fade quickly. The conventional account attributes this to mechanical failure of the particles themselves. Careful in-situ measurements suggest a different picture.

During the first charge, a small amount of the electrolyte decomposes at the negative electrode and builds a passivating crust. Once the crust covers the surface it blocks the reaction that created it, which is why the process is self-limiting under normal conditions. In a Li-ion cell with a graphite anode this is the end of the story.

2. Experimental

Half cells were assembled in an argon-filled glovebox with an oxygen level below 0.5 ppm. Electrodes were cycled between 0.05 and 1.50 V at a rate of 0.5 C using a battery cycler.

3. Results and Discussion

Capacity declined smoothly over 500 cycles with no abrupt step, which is the signature of a process that repeats a little every cycle rather than a single mechanical failure.

$$C(t) = C_0 \exp(-t / \tau) \qquad (1)$$

Figure 1. Discharge capacity versus cycle number for silicon-graphite (circles) and graphite (squares) anodes over 500 cycles at 0.5 C.

Fitting Eq. 1 to the data gives a time constant of 612 cycles. The fit is good ($R^2 = 0.994$), and the residuals show no structure.

Post-mortem imaging supports the interpretation. Particles recovered from cycled electrodes were largely intact, while the surface layer had grown from roughly 20 nm after formation to roughly 140 nm after 500 cycles. If fracture of the particles were the dominant mechanism we would expect the opposite pattern: fragmented particles under a thin, stable layer.

The measured lithium inventory loss accounts for 94 % of the observed capacity fade, leaving little room for a second mechanism of comparable size. Two competing explanations, binder failure and current-collector corrosion, were ruled out by control experiments in which each was suppressed independently without changing the fade rate.

Three practical consequences follow. First, any coating that keeps the interphase intact through a volume change should extend life more than a coating that merely strengthens the particle. Second, cycling protocols that reduce the depth of the volume swing should help, and shallow cycling did indeed reduce the fade rate to 0.0311 % per cycle in a separate series. Third, a lithium reservoir in the cell would postpone rather than prevent the loss.

4. Conclusions

Capacity fade in silicon-graphite anodes is dominated by lithium consumed in rebuilding the interphase, not by particle fracture.

Acknowledgements

This work was supported by the Research Council of Norway under grant 000000. The authors thank the beamline staff.

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